

Nikhil Kotikalapudi

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EDUCATION

The Pennsylvania State University – Schreyer Honors College — School of EECS

State College, PA

Bachelor of Science in Computer Engineering, GPA: 3.94/4.0

Expected Graduation: May 2027

EXPERIENCE

AMD

Austin, TX

Incoming GPU Performance Intern (AI Models)

May 2026

- Designing an automated GPU profiling pipeline for the machine learning frameworks team to identify inference bottlenecks and overcome current profiling-level limitations, using vLLM and kineto traces.
- Optimizing ROCm kernels for AI model performance across GPU architectures.

AlphaGPU

Palo Alto, CA

Software Engineer Intern (Infrastructure)

Aug 2025 – Jan 2026

- Deployed robust GPU profiling for 1000+ users with NVML and torch profiler under Linux, overcoming cloud-hosted GPU limitations with NSYS/NCU.
- Scaled LeetGPU, a platform democratizing hardware-accelerated programming via GPU challenges and online test environments supporting CUDA, Triton, PyTorch, Mojo, and CuTe DSL.

NeuroAI Lab

State College, PA

AI Acceleration Research Assistant

Feb 2025 – Present

- Designed a novel, scalable high-dimensional Bayesian optimization algorithm with dynamic sampling for hyperparameter tuning of neural networks.
- Researching hardware-software co-design of neural networks on neuromorphic and novel architectures, consolidating findings against relevant papers.

IEEE Electronics Packaging Society

Piscataway, NJ

Electronics Research Intern

Feb 2025 – May 2025

- Sustained 96.9% model accuracy under constrained neuromorphic hardware limits (R_ON/R_OFF, bit discretization) via system-level simulation.
- Implemented hardware-in-the-loop training with varying weight quantization, characterizing an 8% accuracy drop on image data.

nth Solutions

Coatesville, PA

Electrical Engineering & Data Analyst Intern

June 2023 – July 2024

- Developed 3 ML-driven signal processing algorithms and benchmarked 100+ nonstationary time-series in Python and MATLAB to reduce vibrations.
- Co-inventor on 2 sensor-based safety products under patent preparation; prototyped a LiDAR ToF device reaching ~90% danger-detection accuracy.

PROJECTS

FlashAttention | CUDA, C++, PyTorch

- Developed a FlashAttention-inspired attention kernel in CUDA achieving 1.54x speedup and 8x more tokens than naïve self-attention, benchmarked against PyTorch SDPA.
- Scaled to 200k+ context on a Tesla T4 with O(N) memory via tiling and online softmax; tuned FP16 tiling and resolved shared-memory bank conflicts to maximize occupancy and HBM throughput.

Neural Network from Scratch, CUDA Accelerated | CUDA, C++, Python, NumPy

- Achieved 10.57x training speedup (40 to 3.8 min) by accelerating a from-scratch deep neural network with a custom CUDA matmul (GEMM) kernel.
- Reached 97.83% on MNIST and 88.02% on Fashion-MNIST with a custom optimizer for dynamic learning-rate scheduling.

4 Essential Machine Learning Projects | Python, PyTorch, Transformers, Scikit-Learn

- Built a RAG bot with the TinyLlama LLM, a pneumonia-detection CNN reaching 90+% accuracy, and a reinforcement learning agent.

TECHNICAL SKILLS

AI/ML & Inference: PyTorch, TensorRT-inspired kernels, vLLM, Transformers, FlashAttention, attention mechanisms, GEMM / matmul kernels, KV-cache, language models (LLMs), multi-modality models, RAG, reinforcement learning, model inference optimization, deep learning

Accelerated Computing & HPC: CUDA, GPU profiling (NSYS/NCU, NVML, torch profiler), Triton, GPU parallelization, FP16 half-precision, memory hierarchy / HBM optimization, occupancy & throughput tuning, ROCm kernels

Languages: Python, C/C++, CUDA, Go, Java, Verilog/SystemVerilog, MIPS Assembly

Tools & Platforms: Linux, Docker, LLVM, NumPy, Scikit-Learn

Core Competency Areas: accelerated computing, AI/ML optimization, HPC, computer architecture, compilers, in-memory computing, signal processing, linear algebra, data analysis